Sujen Kancherla

(916)-896-6761 | sujenk@gmail.com | linkedin.com/in/sujen | github.com/sujen07

Summary

ML engineer with experience building LLM-powered agentic systems and ML pipelines for silicon validation at AMD, and an M.S. in Machine Learning from UCSD. Focused on post-training research in alignment and interpretability.

Technical Skills

Languages: Python, TypeScript, SQL, Verilog

ML: PyTorch, TensorFlow, scikit-learn, XGBoost, HuggingFace, OpenCV, NLTK, LangGraph

Tools: Git, Docker, AWS, Google Cloud, Weights & Biases, ReactJS, NextJS, Vercel AI SDK

Experience

AI/ML Engineer August 2026 – Present
AMD *Folsom, CA*

- Building an **LLM-powered agentic CPU core debug system** to root cause hardware failures; projected to cut time from failure to root cause from ∼**10 days** to ∼**3 hours**
- Building an **LLM agent** that turns historical bug data into targeted test plans and tests
- Drive **AI inference on CPU** validation by turning industry and research trends into new workloads and coverage

AI/ML Engineer Co-op Jan 2025 – May 2026
AMD *Folsom, CA*

- Built an automated ML pipeline that finds failure-producing regression workloads from silicon test data, surfacing ∼**10 RTL bugs per week** with no human intervention
- Built an org-wide RTL bug tracking dashboard, from detection to BIOS patch validation, used by **30+ engineers**
- Built an LLM agent to query live bug and patch status and trigger validation runs on physical silicon

AI Graduate Researcher Jul 2024 – Dec 2024
UCSD ARCLab *La Jolla, CA*

- Trained RL agents (robomimic, NVIDIA Isaac Sim) for autonomous surgical tissue cutting, surpassing established baseline model performance

Machine Learning Intern Jan 2023 – Jun 2023
Omnisync INC *La Jolla, CA*

- Built an end-to-end semantic search pipeline — **live in production** — indexing **7M+ records** using text embeddings and vector search to enable natural language querying over structured data

Data Science Intern Jun 2022 – Sep 2022
Intel *Folsom, CA*

- Automated GPU performance testing with Python and Bash; analyzed temperature and throughput data across production workloads to surface thermal and performance regressions

Education

University of California, San Diego La Jolla, CA
M.S. Machine Learning, Electrical Engineering (Jun 2026) | *B.S. Data Science (2024)*

Publications

- **Avoiding Decision Fatigue with AI-Assisted Decision-Making** — ACM, 2024. Co-authored research on reducing cognitive overload in decision workflows using AI assistance.

Notable Projects

NanoGPT Loss Function & Early Stopping Analysis 2024
LLM Training | *Python, PyTorch, Weights & Biases*

- Trained GPT-2 (124M) on TinyStories and compared alternative loss functions (e.g., Squentropy) and early-stopping strategies against the cross-entropy baseline, evaluating with validation loss and perplexity | GitHub

Paragon AI Calendar Assistant Jun 2025 – Present
Agentic AI | *NextJS, TypeScript, SQL, Vercel AI SDK*

- Production calendar app with an LLM-powered scheduling assistant and full CRUD API layer | Live App

More projects: github.com/sujen07